

Machine Learning- Based Diabetes Risk Prediction System with Streamlit Deployment

Minor Project Report

Submitted by

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Submitted to

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July 20,2026

Abstract

Diabetes is one of the fastest-growing chronic diseases worldwide, making early diagnosis and timely intervention essential for reducing long-term health complications. This project presents a **Machine Learning-Based Diabetes Risk Prediction System with Streamlit Deployment**, designed to estimate an individual's likelihood of developing diabetes using clinical and lifestyle-related parameters.

The system follows a complete end-to-end machine learning workflow consisting of data preprocessing, feature engineering, feature scaling, model training, performance evaluation, and deployment. The dataset is cleaned and prepared before training a Random Forest classifier to identify patterns associated with diabetes risk. Model performance is evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC.

The trained model is deployed as an interactive web application using the Streamlit framework. Users can enter health-related parameters such as Body Mass Index (BMI), blood pressure, cholesterol level, fasting glucose, postprandial glucose, and physical activity. Based on these inputs, the application predicts whether the individual has a high or low risk of diabetes and displays the result instantly.

This project demonstrates the practical application of machine learning in healthcare by integrating data preprocessing, predictive modeling, performance evaluation, and deployment into a single solution. While the application is not intended to replace professional medical diagnosis, it serves as an effective decision-support tool for preliminary diabetes risk assessment and showcases an end-to-end machine learning workflow suitable for real-world deployment.

Keywords: Machine Learning, Diabetes Prediction, Random Forest, Streamlit, Healthcare Analytics, Predictive Modeling.

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Project Overview

The **Machine Learning-Based Diabetes Risk Prediction System** is an end-to-end healthcare analytics application that predicts an individual's diabetes risk using supervised machine learning techniques. The project integrates data preprocessing, model training, performance evaluation, and web deployment into a single workflow, demonstrating the practical application of machine learning in healthcare.

The system is developed using a diabetes dataset containing demographic, clinical, and lifestyle-related attributes. The dataset is preprocessed by handling missing values, preparing features, and applying feature scaling to numerical variables. After preprocessing, the data is divided into training and testing sets for model development. A **Random Forest** classifier is used due to its robustness, ability to handle complex relationships, and strong performance in classification tasks. The trained model is evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC to assess its predictive performance.

To enable real-time prediction, the trained model is deployed as an interactive web application using the **Streamlit** framework. Users can provide health parameters including Body Mass Index (BMI), physical activity, blood pressure, cholesterol level, fasting glucose, and postprandial glucose. These inputs are processed through the same preprocessing pipeline used during training, and the application instantly predicts whether the user is at a high or low risk of diabetes.

Overall, the project demonstrates the complete lifecycle of a machine learning solution, from data preparation and model development to deployment through a user-friendly interface. It highlights how predictive analytics can support early diabetes screening and serves as a practical example of applying Python, Scikit-learn, and Streamlit to develop a deployable healthcare application.

Problem Statement

Diabetes is one of the leading chronic diseases worldwide and has become a major public health concern due to its increasing prevalence and long-term complications. Early identification of individuals at risk is essential for preventing severe health conditions such as cardiovascular diseases, kidney failure, vision impairment, and nerve damage. However, diagnosing diabetes requires the analysis of multiple clinical and lifestyle-related parameters, making the screening process both time-consuming and highly dependent on medical expertise.

Traditional diagnostic approaches primarily rely on laboratory investigations and clinical assessment. Although these methods provide reliable results, they often require specialized equipment, trained healthcare professionals, and considerable processing time. Consequently, large-scale preliminary screening becomes challenging, particularly in remote or resource-constrained regions where timely access to healthcare facilities may be limited.

Machine learning provides an effective solution by analysing multiple health indicators simultaneously and identifying complex relationships within medical data. The objective of this project is to develop an end-to-end machine learning system capable of predicting diabetes risk using important clinical and lifestyle-related parameters. The proposed solution incorporates data preprocessing, feature scaling, model training, performance evaluation, and deployment within a single workflow. A Random Forest classifier is employed due to its robustness, ability to handle complex feature interactions, and strong performance on classification tasks, enabling accurate and reliable diabetes risk prediction.

To make the predictive model accessible to end users, the trained model is deployed as an interactive web application using the Streamlit framework. Users can provide health-related parameters such as Body Mass Index (BMI), physical activity level, systolic and diastolic blood pressure, cholesterol level, fasting glucose, and postprandial glucose. The application processes these inputs through the same preprocessing pipeline used during training and provides an instant prediction indicating whether the individual is at high or low risk of diabetes.

The proposed system is intended to function as a preliminary decision-support tool rather than a replacement for professional medical diagnosis. By integrating machine learning with an intuitive web interface, the project demonstrates a practical, scalable, and user-friendly approach for early diabetes risk assessment while showcasing the complete lifecycle of a deployable machine learning application.

Objectives

The primary objectives of the proposed **Machine Learning-Based Diabetes Risk Prediction System** are as follows:

1. To develop an intelligent machine learning system capable of predicting an individual's diabetes risk using clinical and lifestyle-related health parameters.
2. To collect and preprocess the diabetes dataset by handling missing values, preparing the input features, and ensuring that the data is suitable for machine learning model development.
3. To perform feature scaling on numerical attributes to maintain consistency between the training and prediction phases, thereby improving the overall reliability of the predictive model.
4. To build and train a robust **Random Forest** classification model capable of learning complex relationships between multiple health indicators and diabetes outcomes.
5. To evaluate the trained model using standard machine learning performance metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC in order to assess its predictive capability.
6. To serialize the trained machine learning model and preprocessing components so that they can be reused efficiently during real-time prediction without requiring retraining.
7. To design and develop an interactive web application using the **Streamlit** framework that enables users to enter relevant health parameters and obtain instant diabetes risk predictions.
8. To ensure that user inputs are processed through the same preprocessing pipeline used during model training, thereby maintaining prediction consistency and reducing inference errors.
9. To provide a simple, responsive, and user-friendly interface that can be used by both technical and non-technical users for preliminary diabetes risk assessment.
10. To demonstrate the complete end-to-end lifecycle of a machine learning application, including data preprocessing, model development, evaluation, deployment, and user interaction, while showcasing practical skills in Python, Scikit-learn, and Streamlit.

Technology Stack



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Figure 4.1: Technologies used in the development of the proposed Machine Learning-Based Diabetes Prediction System.

Dataset Description

The performance of any machine learning model largely depends on the quality and relevance of the dataset used during training. For this project, a structured diabetes dataset containing demographic, clinical, and lifestyle-related attributes was used to develop the prediction model. The dataset consists of numerical features that describe an individual's health condition and serves as the primary input for the machine learning pipeline.

The dataset includes important health indicators such as age, Body Mass Index (BMI), physical activity level, systolic blood pressure, diastolic blood pressure, total cholesterol, fasting glucose, postprandial glucose, insulin level, HbA1c, triglycerides, heart rate, and other medical measurements that contribute to diabetes risk assessment. Each record represents a single individual, while the target variable indicates whether the person is diabetic or non-diabetic. These features provide sufficient information for the model to learn the relationship between various physiological parameters and diabetes risk.

Before model development, the dataset was carefully examined to ensure its suitability for machine learning. Numerical attributes were processed using feature scaling to eliminate differences in value ranges and improve model stability. The dataset was then divided into training and testing subsets, enabling the predictive model to learn from one portion of the data while evaluating its performance on previously unseen samples. This approach helps assess the generalization capability of the trained model and reduces the likelihood of overfitting.

Table 5.1 summarizes the key features used in the diabetes prediction system.

Table 5.1: Important Features Used in the Dataset

Feature	Description
Age	Age of the individual (years)
BMI	Body Mass Index
Physical Activity	Weekly physical activity duration
Systolic BP	Systolic blood pressure
Diastolic BP	Diastolic blood pressure
Total Cholesterol	Total cholesterol level
Fasting Glucose	Blood glucose level before meals
Postprandial Glucose	Blood glucose level after meals
Target Variable	Diabetes Status (0 = Non-Diabetic, 1 = Diabetic)

The selected dataset provides a comprehensive representation of the physiological and

lifestyle factors associated with diabetes, making it suitable for developing a reliable machine learning model. The availability of multiple health-related features enables the Random Forest classifier to identify complex relationships among variables and generate accurate predictions for diabetes risk assessment.

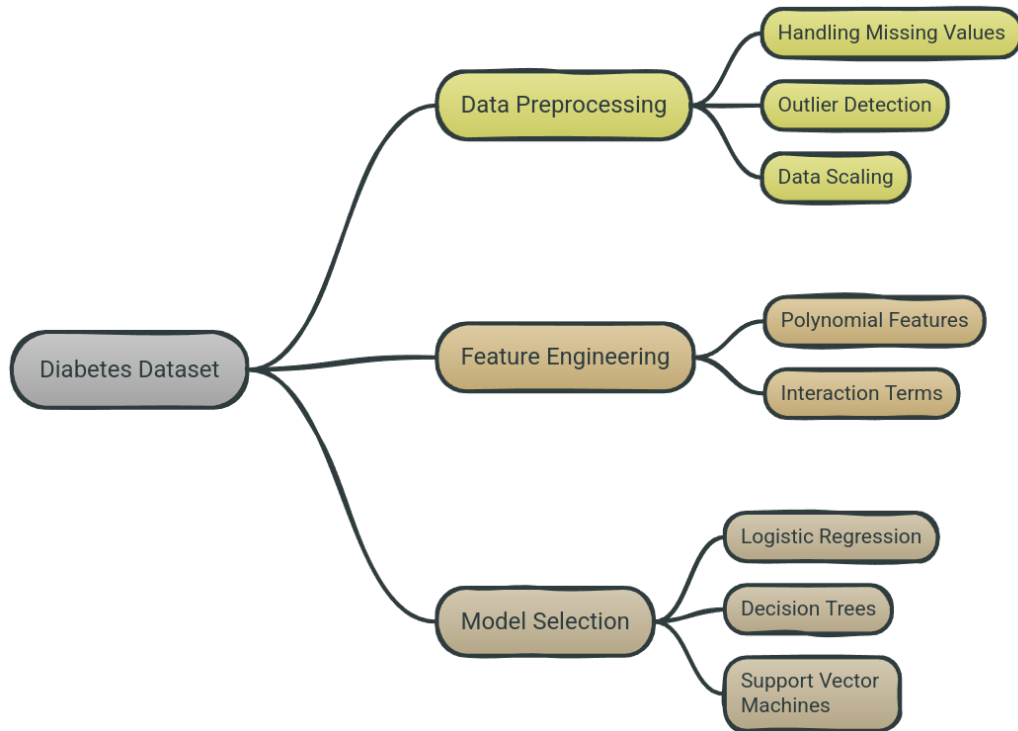
System Architecture

The proposed **Machine Learning-Based Diabetes Risk Prediction System** follows a modular architecture that integrates data preprocessing, machine learning, model deployment, and user interaction into a unified workflow. The system is designed to maintain consistency between the training and prediction phases while ensuring scalability, reliability, and ease of use. The overall architecture consists of three major layers: the data processing layer, the machine learning layer, and the deployment layer, each responsible for a specific stage of the prediction process.

Initially, the diabetes dataset undergoes preprocessing, where the input features are prepared and standardized using the **StandardScaler** technique. The processed data is then divided into training and testing datasets for model development. A **Random Forest Classifier** is trained to learn the relationship between multiple health indicators and diabetes outcomes. After successful evaluation, the trained model, feature scaler, and feature metadata are serialized using the Pickle library, enabling efficient deployment without retraining the model.

The deployment layer is implemented using the **Streamlit** framework, providing an interactive web interface for users. Health parameters entered by the user are processed through the same preprocessing pipeline used during training before being passed to the trained Random Forest model for inference. The system then predicts whether the individual is at a high or low risk of diabetes and displays the result instantly. This modular architecture ensures efficient real-time prediction while allowing future enhancements to be incorporated without affecting the overall system functionality.

Diabetes Dataset Analysis Process



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Figure 6.1: System architecture of the proposed machine learning-based diabetes prediction system.

Methodology

The proposed diabetes prediction system follows a structured machine learning methodology that transforms raw healthcare data into a deployable predictive application. The workflow begins with the collection of a structured diabetes dataset containing demographic, clinical, and lifestyle-related features. The dataset is preprocessed by cleaning the data, selecting relevant features, and applying **StandardScaler** to normalize numerical attributes. The processed dataset is then divided into training and testing subsets to ensure that the predictive model is evaluated on previously unseen data.

A **Random Forest Classifier** is trained using the preprocessed training dataset to learn the relationship between health parameters and diabetes outcomes. The trained model is evaluated using standard classification metrics to assess its predictive performance. After successful evaluation, the trained model, feature scaler, and feature column information are serialized using the Pickle library for efficient deployment. Finally, these serialized components are integrated into a **Streamlit** web application, enabling users to enter their health parameters and receive an instant prediction indicating whether they are at a high or low risk of diabetes. The complete methodology adopted for the proposed system is illustrated in Figure 7.1.

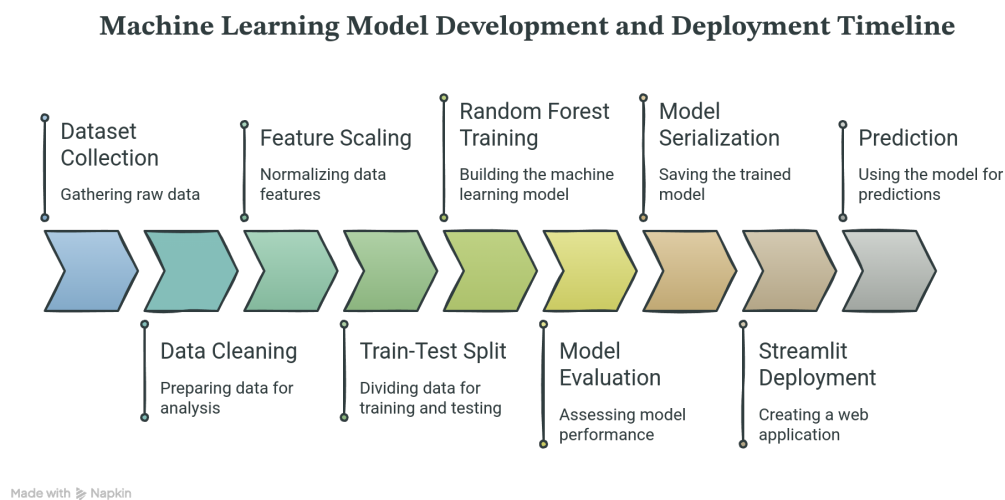


Figure 7.1: Methodology adopted for the proposed diabetes prediction system.

Data Preprocessing

Data preprocessing is an essential step in the machine learning pipeline, as it improves the quality of the dataset and prepares it for effective model training. In this project, the diabetes dataset undergoes several preprocessing steps to ensure consistency, reliability, and better predictive performance. The major preprocessing steps are described below.

1. Data Loading

The diabetes dataset is loaded into the Python environment using the Pandas library. The dataset is stored as a DataFrame, which provides efficient data manipulation and analysis capabilities throughout the preprocessing stage.

2. Data Inspection

The loaded dataset is examined to understand its structure and characteristics. This includes checking the number of records, feature names, data types, and identifying missing or inconsistent values. Initial exploration helps in understanding the quality of the dataset before model development.

3. Feature Selection

The relevant health-related attributes are selected as input features, while the diabetes status is chosen as the target variable. Selecting only the necessary features reduces unnecessary complexity and ensures that the model is trained using meaningful information.

4. Feature Scaling

The numerical features are standardized using the **StandardScaler** technique. Since different health parameters have different numerical ranges, feature scaling ensures that all features contribute uniformly during model training and improves the stability of the predictive model. The fitted scaler is preserved and reused during deployment to maintain consistency.

5. Train-Test Split

The preprocessed dataset is divided into training and testing subsets. The training data is used to develop the Random Forest classifier, while the testing data is reserved for evaluating the model's performance on previously unseen data. This approach helps assess the generalization capability of the model and minimizes overfitting.

Model Development

The model development phase focuses on building a reliable machine learning model capable of accurately predicting an individual's diabetes risk based on selected clinical and lifestyle-related parameters. After completing the preprocessing stage, the prepared dataset is used to train and validate the predictive model. The objective of this phase is to learn the relationship between the input features and the corresponding diabetes outcome while ensuring good generalization on unseen data.

1. Model Selection

The **Random Forest Classifier** was selected as the prediction model for this project due to its high classification performance, robustness, and ability to handle complex relationships among multiple features. Random Forest combines the predictions of multiple decision trees, reducing the risk of overfitting while improving prediction accuracy. It is well suited for healthcare datasets where several clinical parameters collectively influence the target outcome.

2. Model Training

The preprocessed dataset is divided into training and testing subsets before model development. The training dataset is used to build the Random Forest model by learning patterns between the selected health parameters and diabetes status. During training, multiple decision trees are constructed, and the final prediction is obtained through majority voting among all trees, resulting in a stable and reliable classification model.

3. Model Evaluation

After training, the developed model is evaluated using the testing dataset to measure its predictive capability. Standard classification metrics including Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC are used to assess the effectiveness of the model. These metrics provide a comprehensive evaluation of the classifier and help determine its ability to correctly identify diabetic and non-diabetic individuals.

4. Model Serialization

To support real-time deployment, the trained Random Forest model is serialized using the Pickle library. In addition to the trained model, the fitted **StandardScaler** and feature column information are also saved as separate **.pkl** files. These serialized objects are loaded by the Streamlit application during prediction, ensuring that incoming user data undergoes the same preprocessing steps as the training data. This approach eliminates the need for retraining the model and enables efficient, consistent, and scalable diabetes risk prediction.

Streamlit Web Application



Diabetes prediction

Streamlit Diabetes Authenticator ML App

BMI

32.20 - +

Phy Act Mins Per Week

60 - +

systolic_bp

148 - +

diastolic_bp

92 - +

cholesterol_total

245 - +

glucose_fasting

165 - +

glucose_postprandial

265 - +

Predict

You have high risk of diabetes. Please consult a healthcare professional.

Figure 10.1: Streamlit web application showing the user interface and diabetes prediction result.

The final phase of the proposed system involves deploying the trained machine learning model as an interactive web application using the **Streamlit** framework. Streamlit provides a lightweight and efficient platform for transforming machine learning models into user-friendly web applications without requiring extensive web development knowledge. The deployment enables users to interact with the prediction system through a graphical interface rather than executing Python scripts manually.

The application loads the serialized machine learning components, including the trained **Random Forest** model, the fitted **StandardScaler**, and the feature column information, which were previously saved using the Pickle library. Loading these components during application startup eliminates the need to retrain the model each time the application is executed, resulting in faster response times and efficient resource utilization.

The user interface is designed to be simple, intuitive, and easy to use. Users are required to enter relevant health parameters such as Body Mass Index (BMI), physical activity level, systolic blood pressure, diastolic blood pressure, total cholesterol, fasting glucose, postprandial glucose, and other required clinical measurements. Before prediction, the entered values are automatically arranged according to the feature order used during model training and are standardized using the saved scaler. This ensures that the input data follows the same preprocessing pipeline as the training dataset, thereby maintaining prediction consistency.

After preprocessing, the prepared input is passed to the trained Random Forest model, which analyses the health parameters and predicts whether the individual is at a high or low risk of diabetes. The prediction result is displayed instantly on the web interface, allowing users to obtain real-time diabetes risk assessment. The application is designed to provide a smooth user experience while minimizing the complexity of interacting with the underlying machine learning model.

The Streamlit-based deployment demonstrates the practical integration of machine learning and web technologies into a complete end-to-end healthcare application. By combining an interactive interface with a trained predictive model, the proposed system enables rapid diabetes risk prediction and serves as a scalable foundation for future enhancements, including cloud deployment, database integration, and advanced clinical decision-support features.

Experimental Results

The proposed diabetes prediction system was evaluated using the testing dataset after completing the training phase of the Random Forest classifier. The performance of the developed model was assessed using widely accepted classification metrics, including Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and the Confusion Matrix. These metrics provide a comprehensive evaluation of the model's ability to correctly classify diabetic and non-diabetic individuals.

Table 11.1 summarizes the performance achieved by the proposed machine learning model.

Table 11.1: Performance Metrics of the Proposed Random Forest Model

Performance Metric	Obtained Value
Accuracy	99.98%
Precision	100.00%
Recall	99.96%
F1-Score	99.98%
ROC-AUC Score	0.99999

The obtained results demonstrate that the Random Forest classifier effectively learns the relationship between the selected health parameters and diabetes outcomes. The model achieved an overall accuracy of approximately 99.98%, while maintaining excellent precision, recall, and F1-score. The ROC-AUC score, which is very close to 1.0, further indicates the model's strong capability to distinguish between diabetic and non-diabetic individuals across different decision thresholds.

The Confusion Matrix shows that the model correctly classified the majority of test samples, with only a very small number of misclassifications. This indicates that the proposed prediction system is highly reliable for diabetes risk classification. Furthermore, the Feature Importance analysis identifies the most influential health parameters contributing to the prediction, improving the interpretability of the developed machine learning model.

The trained Random Forest model was successfully integrated with the Streamlit web application, enabling users to obtain real-time diabetes risk predictions by entering their health parameters. The experimental results demonstrate that the proposed system provides accurate, efficient, and scalable diabetes risk assessment, making it suitable as a preliminary clinical decision-support tool.

Future Improvements

The proposed Machine Learning-Based Diabetes Prediction System demonstrates excellent predictive performance using the Random Forest classifier and provides an efficient platform for preliminary diabetes risk assessment. Although the developed model achieves outstanding evaluation metrics on the available dataset, there are several opportunities to further enhance the system and improve its applicability in real-world healthcare environments.

1. Validation on Diverse Clinical Datasets

The current model has been trained and evaluated on a structured diabetes dataset and has achieved very high Accuracy, Precision, Recall, and ROC-AUC scores. As a future enhancement, the model can be validated using larger and more diverse clinical datasets collected from multiple hospitals and healthcare organizations. Such validation would further demonstrate the robustness and generalization capability of the proposed system across different patient populations.

2. Expansion of Health Parameters

Additional medical indicators such as family history, dietary habits, smoking status, alcohol consumption, genetic information, medication history, and laboratory test results can be incorporated into the prediction model. Including these features may further improve the comprehensiveness and clinical relevance of diabetes risk assessment.

3. Advanced Machine Learning Techniques

Although the Random Forest classifier provides excellent performance, future work may include evaluating advanced machine learning and deep learning models such as XGBoost, LightGBM, CatBoost, Artificial Neural Networks (ANN), and ensemble learning approaches. A comparative analysis of multiple algorithms can help identify the most suitable model for large-scale healthcare applications.

4. Cloud-Based Deployment

The current Streamlit application can be extended into a cloud-hosted solution using platforms such as AWS, Microsoft Azure, or Google Cloud. Cloud deployment would

enable secure remote access, improved scalability, and support for multiple concurrent users.

5. Integration with Healthcare Systems

The proposed application can be integrated with Electronic Health Record (EHR) systems and hospital management software to automatically retrieve patient information and assist healthcare professionals during clinical decision-making. Such integration would improve workflow efficiency while reducing manual data entry.

6. Enhanced User Experience

Future versions of the application may include personalized health recommendations, risk visualization dashboards, multilingual support, secure user authentication, and mobile-friendly interfaces. These improvements would enhance usability and make the system more accessible to a broader population.

Overall, the proposed system establishes a strong foundation for intelligent diabetes risk prediction. With additional validation, expanded datasets, cloud deployment, and integration into real-world healthcare infrastructure, the system can evolve into a comprehensive clinical decision-support application capable of assisting healthcare professionals and promoting early detection of diabetes.

Conclusion

The increasing prevalence of diabetes highlights the need for intelligent and efficient tools that can assist in the early identification of individuals at risk. In this project, a comprehensive machine learning-based diabetes prediction system was successfully designed, developed, and deployed to provide preliminary diabetes risk assessment using clinical and lifestyle-related health parameters. The proposed system demonstrates how machine learning can be effectively integrated into healthcare applications to support timely decision-making and improve accessibility to predictive analysis.

The project followed a structured development methodology consisting of dataset preparation, data preprocessing, feature scaling, model development, performance evaluation, and deployment through a Streamlit-based web application. The Random Forest classifier was selected as the prediction model due to its robustness, high classification capability, and ability to model complex relationships among multiple health indicators. The trained model, along with the preprocessing components, was serialized using the Pickle library, enabling efficient real-time prediction without repeated model training.

Experimental evaluation demonstrated excellent predictive performance, with an accuracy of approximately **99.98%**, precision of **100.00%**, recall of approximately **99.96%**, and an ROC-AUC score close to **1.0**. These results indicate that the developed model is highly effective in distinguishing between diabetic and non-diabetic individuals within the evaluated dataset. Furthermore, the successful deployment of the model through an interactive Streamlit application confirms its practical applicability and provides users with an intuitive platform for instant diabetes risk prediction.

Beyond achieving strong predictive performance, the project demonstrates the complete lifecycle of a real-world machine learning application, beginning with raw healthcare data and ending with a fully functional web-based prediction system. The modular design of the application allows future enhancements, including the integration of additional clinical parameters, validation using larger healthcare datasets, cloud deployment, and integration with hospital information systems.

In conclusion, the proposed Machine Learning-Based Diabetes Prediction System successfully achieves its primary objective of providing an accurate, efficient, and user-friendly solution for diabetes risk prediction. The project not only demonstrates the practical application of machine learning techniques in the healthcare domain but also showcases the integration of data science, software development, and web deployment into a unified intelligent system. With further validation using diverse real-world clinical datasets, the proposed system has the potential to evolve into a reliable clinical decision-support tool for assisting healthcare professionals in the early detection and management of diabetes.

References

The following references were consulted during the design, development, implementation, and evaluation of the proposed Machine Learning-Based Diabetes Prediction System. These sources include official software documentation, machine learning libraries, and datasets that supported various stages of the project.

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